METRIC

MIL-L-2104E
AMENDMENT 1
26 May 1989

MILITARY SPECIFICATION

LUBRICATING OIL, INTERNAL COMBUSTION ENGINE, TACTICAL SERVICE

This amendment forms a part of MIL-L-2104D, dated 1 August 1988, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

2.2, under AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM), D 2896, delete the word "Total" from the title and add:

"D 4624 - Measuring Apparent Viscosity by Capillary Viscometer at High Temperature and High-Shear Rates.
"D 4741 - Measuring Viscosity at High Temperature and High Shear Rate by Tapered-Plug Viscometer.
"D 4927 - Elemental Analysis of Lubricant and Additive Components - Barium, Calcium, Phosphorus, Sulfur, and Zinc by Wavelength Spectroscopy."

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2.2, add the following at the end of the section:

DEUTSCHES INSTITUT FUR NORMUNG (DIN)

DIN 51581 - Determination of evaporation loss of lubricating oils.

(Application for copies should be addressed to Deutsches Institut fur Normung e.V., Burggratenstr 4-10, D-1000 Berlin 30, Germany T 26011.)

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Table II, after "High temperature/high shear viscosity" insert an "X" in the columns under 10W, 30 and 40 grades.

Under "Viscosity index, (min)", grade 40, delete the "X" and substitute "80".

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4.5.2, add the following tests:

"Evaporation loss
High temperature/high shear".

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Table III, delete in its entirety and substitute the following:

TABLE III. Test methods.

| Test | Test method | | | |
|---|---|---|---|---|
| | FED-STD-791 | ASTM | SAE | DIN |
| Viscosity, kinematic | | D 445 | | |
| Viscosity, apparent[1/] | | | J300 | |
| High temperature/high shear | | D 4683, D 4624, D 4741 | | |
| Viscosity index | | D 2270 | | |
| Pour point | | D 97 | | |
| Stable pour point | 203 | | | |
| Borderline pumping | | D 4684 | | |
| Flash point | | D 92 | | |
| Evaporative loss[2/] | | D 2887 | | 51581 |
| Gravity, API | | D 287 | | |
| Carbon residue | | D 524 | | |
| Color | | D 1500 | | |
| Total acid number | | D 664 | | |
| Base number | | D 2896 | | |
| Phosphorus | | D 1091, D 4047 | | |
| Chlorine | | D 808, D 1317[3/] | | |
| Sulfur[4/] | | D 129, D 1552, D 2622, D 4294 | | |
| Nitrogen | | D 3228 | | |
| Saponification number | | D 94 | | |
| Sulfated residue | | D 874 | | |
| Boiling range distribution | | D 2887 | | |
| Metallic components | 5601 | D 4628[5/], D 4927 | | |
| Foaming | | D 892 | | |
| Stability & compatibility | 3470[6/] | | | |
| Moisture-corrosion characteristics | | Sequence IID[7/] | | |
| Oxidation & wear characteristics | | Sequence IIIE[7/] | | |
| Low temperature deposits & wear | | Sequence VE[7/] | | |
| Bearing corrosion & shear stability | | Labeco L-38[8/] | | |
| Ring-sticking, wear, & accumulations of deposits | | | | |
| Four-stroke cycle diesel engine[8/] | | Caterpillar 1G2 | | |
| Two-stroke cycle diesel engine | 355 | | | |
| Friction retention characteristics & wear: | | | | |
| Slip time & wear | | Allison C-3[9/] | | |
| Stopping time & wear | | D 4736 | | |
| Seal compatibility | | Allison C-3[10/] | | |

1/ Obtain the apparent viscosity using the method of test set forth by appendix A of SAE J300.



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2/ The DIN 51581 is the preferred method.
3/ ASTM D 808 is the preferred method.
4/ ASTM D 1552 is the preferred method. ASTM D 4294 is only for use with base stock.
5/ Other spectrochemical analysis methods as approved by the qualifying activity (see 6.4) may be used as alternates.
6/ See 4.6.1 for clarifying instructions.
7/ In accordance with ASTM STP 315H.
8/ In accordance with STP 509A.
9/ Use procedure described in item 9, Allison Transmission Division (ATD) C-3 specification.
10/ Use procedure described in item 6, Allison Transmission Division (ATD) C-3 specification.

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Add the following at the top of the page:

"4.5 Shear stability. Determine the shear stability of grade 15W-40 oil by the following method:"

Custodians:
Army - ME
Navy - SH
Air Force - 68

Preparing activity:
Army - ME

Project 9150-1045

Review activities:
Army - AR, SM
Navy - AS, MC, SA, YD
Air Force - 11
DLA - GS

User activities:
Army - AT, MI
Navy - OS



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